

Technical Progress Report

Project name:	Biped robot		
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Start date:	2026-03-07	End date:	2026-03-13
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1. Executive Summary

This report updates the second-week progress of the Biped Robot project. Based on week 1's preparatory work, key achievements include confirming the motor layout of the 6-DOF biped robot, formulating the differential speed turning method, and finishing preliminary 3D modeling of mechanical components. The project encountered design and 3D printing modeling problems, and is now in the mechanical design and modeling optimization phase, laying a structural foundation for subsequent part printing and assembly.

2. Objectives & Scope

- Confirm the rational motor layout of the 6-DOF biped robot, matching motor functions with straight walking and steering requirements.
- Formulate a feasible differential speed turning method, and complete preliminary 3D modeling combined with motor layout.
- Sort out problems in design and modeling, and put forward preliminary solutions to lay a foundation for part printing and assembly.

3. Technical Work Completed

3.1 Motor Layout Confirmation

- Determined the 6-DOF robot's motor layout: 2 motors for steering control, 4 for straight walking control, the latter arranged at the thigh and knee joints of both legs.
- Optimized motor installation positions, verified compatibility with selected MG996R/MG90S servos, and ensured no structural interference between motor layout and mechanical structure.

3.2 Formulation of Turning Method

- Confirmed differential speed turning as the core method, realizing turning by controlling the step length difference of left and right legs.
- Clarified the motor control division during turning, completed theoretical verification of the method, and confirmed its feasibility for the 6-DOF structure.

3.3 Preliminary 3D Modeling

- Completed preliminary 3D modeling of key mechanical components (thigh frame, motor fixing bracket, etc.) via SolidWorks based on motor layout and turning method.
- Carried out preliminary virtual assembly, and marked key size parameters of components in the model for subsequent optimization.

3.4 Problem Sorting

- Collected and sorted out project problems, divided them into design problems and 3D printing modeling problems, and put forward preliminary solution ideas and optimization directions.

4. Observations & Findings

- The motor layout (2 for steering, 4 for straight walking) is highly matched with the 6-DOF structure, with clear functional division and reasonable space layout. The differential speed turning method is simple and feasible, and highly compatible with the current mechanical structure.
- The project progress is consistent with the 2-week timeline, the preliminary 3D model basically matches the movement requirements, and key parameters are clear.
- The encountered problems have clear types and targeted solution ideas, which can be solved through local optimization of the design scheme and model adjustment.

5. Challenges & Roadblocks

- Design problems: Unreasonable structural design at the motor-mechanical connection part may reduce installation stability; imperfect leg joint design may affect the coordination of straight walking and turning.
- 3D printing modeling problems: Imprecise partial size parameters of the model may cause part mismatch with motors; the model does not fully consider printing process requirements, which may lead to printing defects.
- The transformation of differential speed turning control logic into actual motor drive code has unknown difficulties and needs further verification.

6. Next Steps & Recommendations

- Optimize the 3D model according to existing problems, adjust inaccurate size parameters, and modify unreasonable structural designs to meet printing and installation requirements.
- Confirm the final model and output formal 3D printing files with clear parameters.
- Carry out 3D printing of all mechanical parts and check the quality and dimensional accuracy of printed parts.
- Conduct the overall assembly of the biped robot, including motor installation, mechanical component connection, and preliminary debugging of part positions.
- Prepare assembly tools and materials in advance, and sort out the assembly sequence to improve efficiency and accuracy.



7. Conclusion

- In the second week, the project made important structural design progress on the basis of week 1's work: confirmed the motor layout of the 6-DOF robot, formulated the differential speed turning method, and completed preliminary 3D modeling of key components. The team also sorted out design and modeling problems and put forward feasible preliminary solutions.